Cool macrofauna title

Jason Berger[1*], Nina Farah[1*], Juliette Kiffer[2*], Nelle Konczak[1*]

Abstract
Enter here our amazing abstract!

Keywords
soil macrofauna, Paracou, Amazon forest, ecology

[1]FTH 2026 student Université de Guyane, UMR EcoFoG, Campus agronomique, Kourou, French Guiana, France, 97310
[2]FTH 2026 student AgroParisTech, UMR EcoFoG, Campus agronomique, Kourou, French Guiana, France, 97310
***Corresponding Author**: jasonberger2002@gmail.com
***Corresponding Author**: ninafarah@free.fr
***Corresponding Author**: jujukiffer95@gmail.com
***Corresponding Author**: konczaknelle@gmail.com

Table of contents

1 Introduction **1**
1.1 Context 1 . . . 1
1.2 Context 2 . . . 1
2 Material and Methods **1**
2.1 Study Site . . . 1
2.2 Sampling and Identification . . . 1
2.3 Statistical analysis . . . 1
Explorative Analysis ▪ Analysis of Abundances (Generalized Linear Mixed Models) ▪ Analysis of Community Composition (PERMANOVA)
3 Results **1**
3.1 General . . . 1
3.2 GLMM . . . 1
Global Abundance ▪ Formicidae ▪ Isoptera ▪ Lumbricina Oligochaetes
3.3 PERMANOVA . . . 3
4 Discussion **3**
5 Conclusion **3**
6 INSTRUCTIONS TEMPLATE, REMOVE LATER **3**
6.1 Lists . . . 3
6.2 Math . . . 3
6.3 Cross-references . . . 5
6.4 Bibliography . . . 5
6.5 Forcing line breaks . . . 5
References **5**

1. Introduction

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

1.1 Context 1

text

1.2 Context 2

text

2. Material and Methods

2.1 Study Site

Refer to Figure 1

PLACEHOLDER citation Ramos et al. (2025) or (Ramos et al., 2025), or Freycon et al. (2003) or Pavlíček & Csuzdi (2012)

2.2 Sampling and Identification

text

2.3 Statistical analysis

negative binomial (NB2) and Poisson tested

Explorative Analysis

text

Analysis of Abundances (Generalized Linear Mixed Models)

text

Analysis of Community Composition (PERMANOVA)

text

3. Results

3.1 General

Densities by soil type and layer Figure 2

Relative composition of community abundances Figure 3

Explorative ordination PCA Figure 4

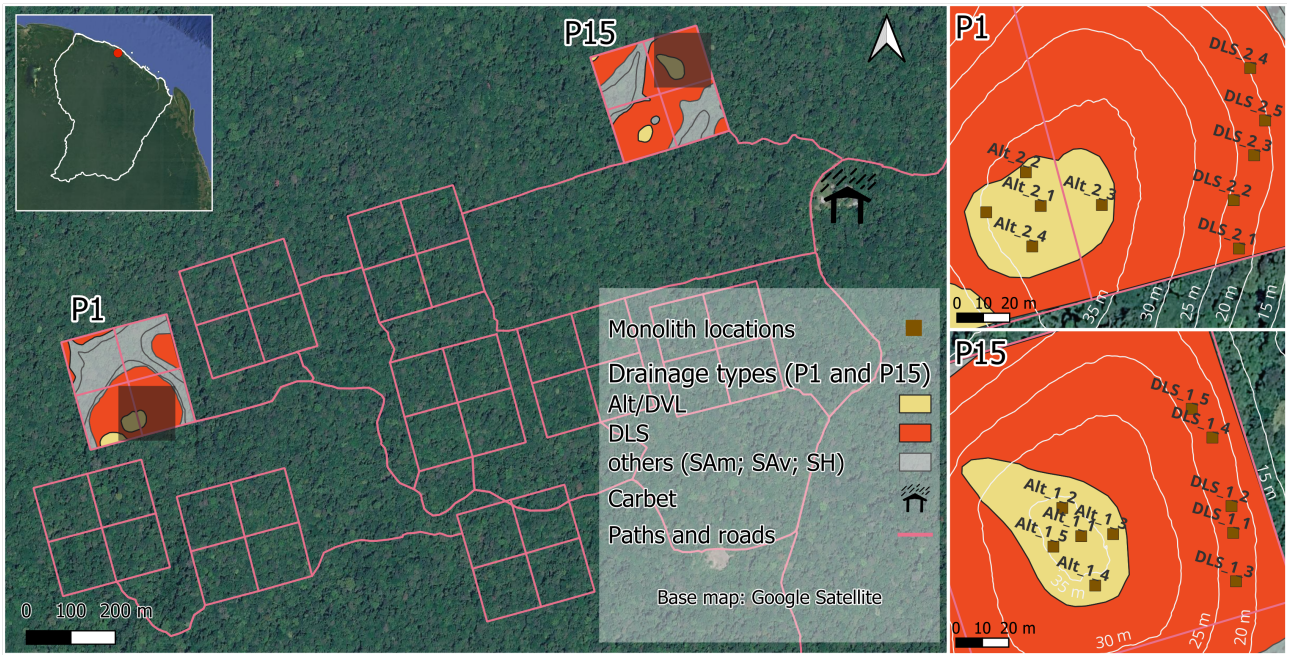


Figure 1. Overview map of Paracou with the sampling locations. Location of right panels shown by grey squares. White lines on the right panels represent altitude isolines.

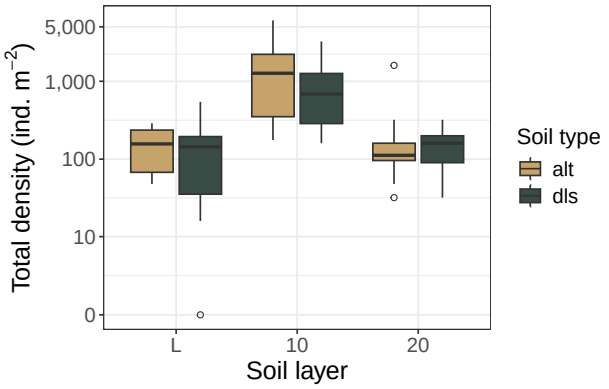


Figure 2. Boxplot with total macrofauna densities by soil type and layer. y-axis is logarithmic.

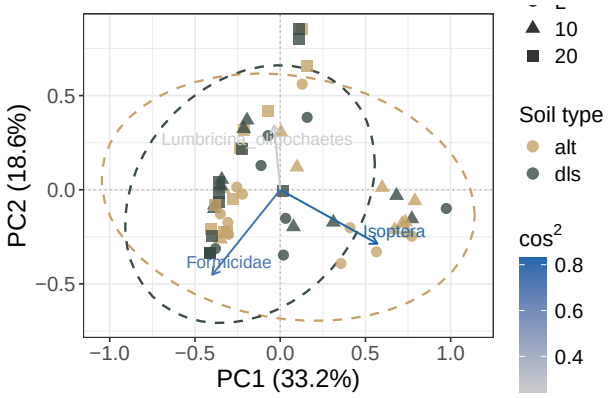


Figure 4. PCA ordination based on Hellinger-transformed community matrix. First two dimensions are presented. Only taxa with $\cos^2 > 0.1$ are presented.

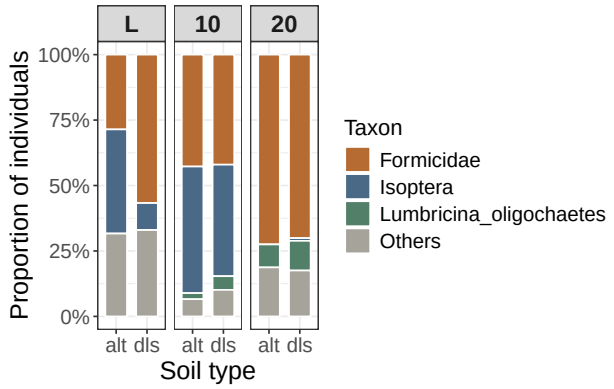


Figure 3. Relative composition of macrofaunal community abundances by soil type and layer.

3.2 GLMM

Effects of soil type and layer were tested on global, ant, termite and earthworm abundances. All retained models described below are listed in Table 2.

Global Abundance

Type II Wald chi-square tests revealed a highly significant effect of soil layer on total soil macrofauna abundance ($\chi^2 = 106.85$, df = 2, $p < 0.001$). The effect of soil type was not significant ($\chi^2 = 0.52$, df = 1, $p = 0.469$). Estimated random-effect variance (monolith) was 0.437 (SD = 0.661), indicating variation in baseline macrofauna abundance among monoliths.

Pairwise comparisons with Tukey adjustment (Table 1) showed that macrofauna abundance was significantly higher in layer 10 than in layers L (7.42 times) and 20 (6.18 times), while no significant difference was found between layers L and 20 ($p = 0.714$).

Formicidae

Type II Wald chi-square tests revealed a highly significant effect of soil layer on ant abundance ($\chi^2 = 28.00$, df = 2, $p < 0.001$). The effect of soil type was not significant ($\chi^2 = 0.48$, df = 1, $p = 0.486$). The estimated random-effect variance for monolith was 0.655 (SD = 0.810), indicating variation in baseline ant abundance among monoliths. Quantile deviation was detected in the DHARMa residuals for the presented model. Specifically, upper quantile deviated significantly and thus relatively high-abundance observations may not be adequately captured by the model. Since the diagnostics did not indicate deviations regarding dispersion, zero-inflation, uniformity as well outliers, and the model has the best AIC support, it was retained, regardless.

Pairwise comparisons with Tukey adjustment (Table 1) showed that ant abundance was significantly higher in layer 10 than in layer L (5.05 times) and layer 20 (3.27 times), while no significant difference was found between layers L and 20 ($p = 0.3995$).

Isoptera

Type II Wald chi-square tests revealed a highly significant effect of soil layer on termite abundance ($\chi^2 = 36.56$, df = 2, $p < 0.001$). The effect of soil type was not significant ($\chi^2 = 0.89$, df = 1, $p = 0.347$). The estimated random-effect variance for monolith was effectively zero (8.22×10^{-9}; SD = 0.00009), indicating virtually no remaining variation in baseline termite abundance among monoliths after accounting for the fixed effects.

Pairwise comparisons with Tukey adjustment (Table 1) showed that termite abundance was significantly higher in layer 10 than in layer L (18.08 times) and layer 20 (745.48 times; estimated richness for 20 nearing zero). Abundance was also significantly higher in layer L than in layer 20 (41.23 times; all p 0.0073).

Lumbricina Oligochaetes

results text

3.3 PERMANOVA

PERMANOVA did not show a significant difference in macrofaunal community composition between soil types ($F = 1.68$, $R^2 = 0.085$, $p = 0.139$). The test for homogeneity of multivariate dispersion was also non-significant ($F = 0.11$, $p = 0.742$), indicating no detectable difference in within-group dispersion between soil types. Layers were aggregated per monolith to account for spatial dependence.

PERMANOVA revealed a significant effect of soil layer on macrofauna community composition (Bray–Curtis dissimilarity: $F = 5.46$, $R^2 = 0.163$, $p = 0.0001$). Soil layer explained around 16.3% of variation in the community composition, indicating that macrofauna community composition differed significantly among layers. Permutation was restricted within monolith to account for spatial dependence of layers. Homogeneity of multivariate dispersion was assessed using a permutation test based on betadisper. No significant differences in dispersion were detected for soil layers ($F = 0.75$, $p = 0.321$), suggesting that the PERMANOVA result was not driven by differences in within-group dispersion.

Pairwise PERMANOVA suggests significant differences in macrofauna community composition for all soil layers (Table 3). Significant differences were observed between L and 10 ($F = 6.05$, $R^2 = 0.141$, adjusted $p = 0.0003$), L and 20 ($F = 4.55$, $R^2 = 0.110$, adjusted $p = 0.0003$), and 10 and 20 ($F = 5.76$, $R^2 = 0.132$, adjusted $p = 0.0012$), using Bonferroni correction.

4. Discussion

text

5. Conclusion

text

6. INSTRUCTIONS TEMPLATE, REMOVE LATER

6.1 Lists

Lists are indicated by `*`, `+` and `-` (three hierarchical levels) or numbers `1.`, `i.` and `A.` (numbered lists). Indentation of lists indicates their level: `*`, `+` and `-` may be replaced by `-` at all levels, but four spaces are needed to nest a list into another.

- First element of a list
    - sub-list

Table 1. Pairwise emmeans comparisons of abundance by layer for total macrofauna, ants (Formicidae) and termites (Isoptera).

Taxon	Comparison	Abundance ratio	z-ratio	Adjusted p-value
Total macrofauna	L - 10	0.13	-9.12	<0.0001***
Total macrofauna	L - 20	0.83	-0.78	0.713689
Total macrofauna	10 - 20	6.18	8.34	<0.0001***
Ants (Formicidae)	L - 10	0.20	-4.96	<0.0001***
Ants (Formicidae)	L - 20	0.65	-1.29	0.399482
Ants (Formicidae)	10 - 20	3.27	3.84	0.000363***
Termites (Isoptera)	L - 10	0.06	-4.06	0.000145***
Termites (Isoptera)	L - 20	41.23	3.01	0.007262**
Termites (Isoptera)	10 - 20	745.48	5.42	<0.0001***

Abundance ratios are exponentiated log-scale contrasts. Significance levels: * p < 0.05; ** p < 0.01; *** p < 0.001. P-values adjusted using the Tukey method within each taxon.

Table 2. Summary of retained GLMMs for global abundance, ants (Formicidae), termites (Isoptera) and earthworms (Lumbricina oligochaetes). ADD MORE DESCRIPTION HERE

Response	Distribution	Predictor	Chi-square	df	p-value
Total macrofauna	Negative binomial	soil_type	0.52	1	0.46933
Total macrofauna	Negative binomial	layer	106.85	2	<0.0001***
Ants	Negative binomial	soil_type	0.48	1	0.48633
Ants	Negative binomial	layer	28.00	2	<0.0001***
Termites	Negative binomial	soil_type	0.89	1	0.34664
Termites	Negative binomial	layer	36.56	2	<0.0001***
Earthworms	Poisson	soil_type	0.35	1	0.55692
Earthworms	Poisson	layer	12.69	2	0.00176**

Type II Wald chi-square tests. All models included soil type and soil layer as fixed effects and monolith ID as a random intercept. Significance levels: * p < 0.05; ** p < 0.01; *** p < 0.001

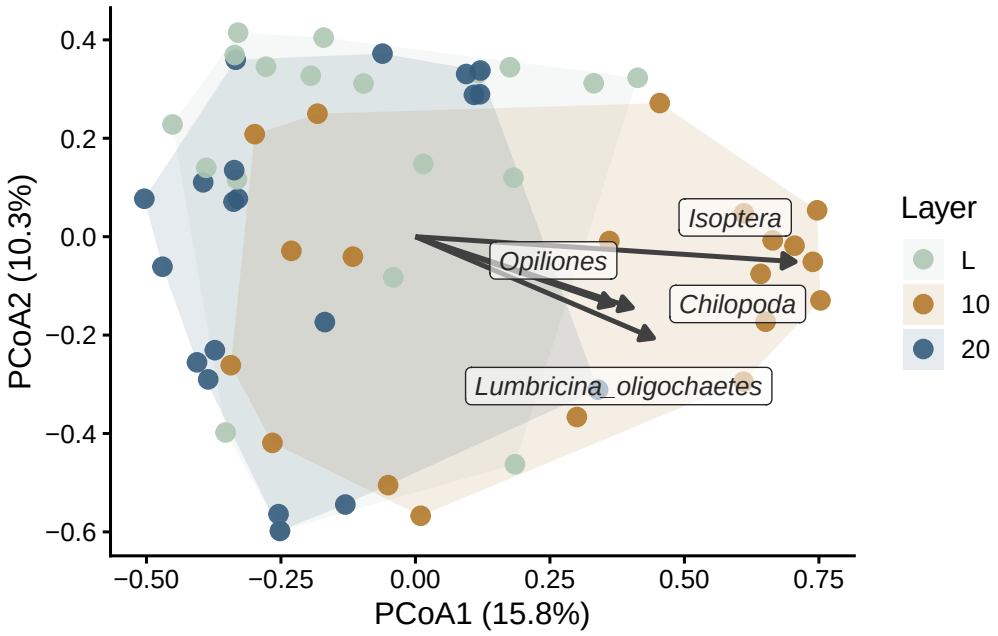


Figure 5. Bray-Curtis PCoA for layers here

Table 3. Pairwise PERMANOVA for community composition comparisons between layers.

Comparison	F-statistic	R^2	Adjusted p-value
L - 10	6.05	0.141	0.0003***
L - 20	4.55	0.110	0.0003***
10 - 20	5.76	0.132	0.0021**

Significance levels: * p < 0.05; ** p < 0.01; *** p < 0.001; p-values adjusted using Bonferroni method

- Second element
- Continuation of the list

Leave an empty line before and after the list, but not between its items.

6.2 Math

Equations in LaTeX format can be inserted inline, like $A = \pi r^2$ or isolated like

$$e^{i\pi} = -1.$$

They can be numbered, see Equation 1, after adding them a label:

$$A = \pi r^2. \tag{1}$$

6.3 Cross-references

Figures and tables have a label declared in their division `tbl-xxx` or `fig-yyy`, starting with `fig-` or `tbl-`.

For equations, the label is added manually by the code `{#eq-xxx}` after the end of the equation.

Sections can be tagged by ending their title with `{#sec-yyy}`.

In all cases, the call to the reference is made by `@`.

6.4 Bibliography

Bibliographic references included in the `.bib` file declared in the document header can be called by `[@CitationKey]`, in parentheses (Ramos et al., 2025), or without square brackets, in the text, as Ramos et al. (2025).

The bibliographic style can be specified, by adding the line

```
csl: file_name.csl
```

in the document header and copying the *.csl* style file to the project folder. More than a thousand styles are available[1] and their URL can be used instead of copying the file, e.g.:

```
csl: https://www.zotero.org/styles/xxx
```

6.5 Forcing line breaks

Hyphenation is handled automatically in LaTeX. If a word is not hyphenated correctly, add its hyphenation in the preamble of the file with the command `hyphenation` (words are separated by spaces, hyphenation locations are represented by dashes).

If LaTeX can't find a solution for the line break, for example because some code is too long a non-breaking block, add the LaTeX command `\break` to the line break location. Do not leave a space before the command. The HTML document ignores LaTeX commands.

References

Freycon, V., Sabatier, D., Paget, D., & Ferry, B. (2003). Influence du sol sur la végétation arborescente en forêt guyanaise : État des connaissances. *Revue Forestière Française*, (55 (sp)), 60–73. https://doi.org/10.4267/2042/5787

Pavlíček, T., & Csuzdi, C. (2012). Earthworm fauna of French Guiana: (Oligochaeta). *Zoology in the Middle East*, *58*(sup4), 107–110. https://doi.org/10.1080/09397140.2012.10648991

Ramos, A. P. D., Lima, S. S. D., Ferreira, C. D. S., Pinto, L. A. R. D. S., Ferreira, R., Dias, A., Matos, P. S., & Pereira, M. G. (2025). Macrofauna and soil properties in agroforestry system and secondary forest. *Revista Brasileira de Ciência do Solo*, *49*, e0240091. https://doi.org/10.36783/18069657rbcs20240091

[1] https://github.com/citation-style-language/styles